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Mobile Clinic Social Needs Screening — QI Project

An academic quality improvement (QI) project for a mobile clinic in **Chantilly, Virginia (Fairfax County)** serving about 30 patients experiencing homelessness.

Problem: Social needs screening depends on which staff member sees the patient. Some patients are asked about food, housing, or transportation, and others are never asked.

Goal: Every eligible patient is screened with the same standardized questionnaire, and every identified need is offered a referral with documented follow-up.

> **Phase 1 (this repository):** research, screening protocol, implementation plan, measurement plan, data templates, and analysis code. There is no application code yet. **Phase 2:** decide what to build. See docs/09-phase2-decision-guide.md.

Contents

#	Document	Purpose
00	Project charter	Problem, SMART aim, scope, team, timeline
01	Evidence review	Screening tool comparison and recommendation
02	Screening questionnaire	The standardized instrument and scoring rules
03	Screening protocol (SOP)	Who, when, how; positive-screen response
04	Resource directory	Fairfax County / Chantilly referral resources
05	Training plan	Staff curriculum, scripts, competency check
06	Implementation plan	Phases, PDSA cycles, driver diagram, roles
07	Measurement plan	Measure definitions, run-chart rules, targets
08	Ethics, privacy & QI determination	IRB/QI determination, HIPAA, safety reporting
09	Phase 2 decision guide	Open questions and build options
10	Real data plan	Getting real data: approvals, public sources, contacts, outreach templates
—	Data dictionary	Field definitions for the CSV logs
—	Templates	Printable screening form, PDSA worksheet, logs

Data analysis

The data templates and the synthetic dataset use the same schema, so the analysis runs unchanged on real data later.

```
pip install -r analysis/requirements.txt
python analysis/generate_synthetic_data.py --weeks 16 --baseline-weeks 4 --seed 42
python analysis/analyze.py --data-dir data/synthetic --out-dir output
```

Outputs: run charts in `output/figures/` and a measure summary in `output/summary.md`.

> ⚠️ Everything in `data/synthetic/` is **simulated**. It shows how the analysis works and must not be cited as findings.

Real data goes in `data/real/`, which is git-ignored, and only on clinic-approved storage. See 10 first. Keep `data/synthetic/` and `output/` as the demo.

```
python analysis/analyze.py --data-dir data/real --out-dir data/real/output --strict
```

Word / PDF versions

```
./scripts/build_docs.sh     # requires pandoc; writes dist/*.docx and dist/*.pdf
```

Key decisions so far

- **Instrument:** AHC-HRSN core items, adapted for homelessness, plus the Single Item Literacy Screener and a locally developed ID/documents item. See 01.
- **Domains:** housing, food, transportation, utilities, interpersonal safety, health literacy, ID/documents.
- **Out of scope:** behavioral health and substance-use screening stay in the separate clinical workflow until referral capacity is confirmed.
- **Timeline:** not decided. The plan uses a 16-week default (4 baseline + 12 intervention), which can be changed.
- **Data system:** unknown. The plan is paper-first and system-agnostic.

00 — Project Charter

Project title

Standardizing Social Needs Screening and Referral at a Mobile Clinic Serving People Experiencing Homelessness (Chantilly, VA)

Problem statement

The mobile clinic serves about 30 adults experiencing homelessness in western Fairfax County. Staff do not use a standard way to ask about social needs. Whether a patient is asked about food, transportation, safety, or documents depends on the clinician. Positive answers are not documented in a consistent place, and referral follow-up is not tracked. As a result, needs go unidentified, referrals are inconsistent, and the clinic cannot tell whether patients are connected to services.

Background

- Unmet health-related social needs (HRSNs) are associated with worse health outcomes and higher acute-care use. For people experiencing homelessness, several HRSNs often occur together.
- Standardized instruments exist, such as CMS AHC-HRSN and NACHC PRAPARE. Their value depends on a reliable workflow and a referral pathway the clinic can actually deliver (see 01).

SMART aim (global)

> By the end of the 12-week intervention period, **≥ 80 %** of adult patient visits will have an up-to-date (completed within 90 days) standardized social needs screen (baseline estimated < 30 %), and **≥ 90 %** of patients with a positive screen will be offered a referral for every positive domain at the same visit.

Secondary aim: **≥ 50 %** of accepted referrals will have a documented follow-up outcome within 30 days.

Targets are provisional. Replace the baseline estimate after the baseline chart audit (07).

Scope

In scope	Out of scope (for now)
Adults (≥ 18) seen at the mobile clinic	Pediatric screening
Domains: housing, food, transportation, utilities, interpersonal safety, health literacy, ID/documents	Behavioral health and substance-use screening (separate clinical workflow)
Staff training, screening workflow, referral offer, follow-up tracking	Building software (Phase 2 decision)
Fairfax County / Chantilly resources	Direct provision of services

Team (fill in names)

Role	Responsibility	Name
Project lead (student)	Planning, data analysis, write-up	
Faculty advisor / preceptor	Academic oversight, QI determination	
Clinic sponsor (medical director / manager)	Approval, removing barriers	
Screening champion (RN / MA / CHW)	Day-to-day workflow, peer coaching	
Referral/follow-up owner (CHW / case manager)	Warm handoffs, 30-day follow-up	
Patient/lived-experience advisor	Wording and acceptability review	

Timeline (default 16 weeks — adjust when decided)

Weeks	Phase
−4 to 0	Preparation: approvals, QI/IRB determination, resource verification, training
1–4	Baseline: retrospective/concurrent chart audit, no change in practice
5–8	PDSA cycle 1: paper screening at intake by one champion
9–12	PDSA cycle 2: spread to all staff + referral offer workflow
13–16	PDSA cycle 3: 30-day follow-up calls / closed-loop tracking
17+	Sustainment plan, final analysis, SQUIRE 2.0 write-up

Measures (summary — see 07)

- **Process:** % visits with up-to-date screen; % due screens completed; % positive screens offered referral.
- **Outcome:** % accepted referrals with confirmed connection within 30 days.
- **Balancing:** added visit time; patient decline rate; staff-reported burden.

Risks and mitigations

Risk	Mitigation
Screening for needs the clinic cannot address erodes trust	Verified resource directory before go-live; ask whether the patient wants help
Safety disclosures in a non-private mobile space	Ask safety items only when privacy is possible; script for deferral
Patients hard to reach for follow-up	Collect best contact method; follow up at the next clinic visit
Small sample (n≈30) limits statistical inference	Use run charts and run rules, not hypothesis tests
Staff turnover/volunteers	Brief job aid, champion model, training in onboarding

Approvals

- [ ] Clinic sponsor sign-off
- [ ] Academic program / faculty sign-off
- [ ] QI vs. research determination (08)

01 — Evidence Review: Selecting a Social Needs Screening Instrument

Question

Which standardized instrument should a small mobile clinic serving adults experiencing homelessness use? It should cover housing, food, transportation, utilities, interpersonal safety, health literacy, and ID/documents. It must be short enough for a mobile visit, and every domain must map to a referral the clinic can actually make.

Method

Targeted review of instrument documentation, the UCSF SIREN screening-tool comparison (which also covers WE CARE, Health Leads, and others), and key primary literature (references below). This is not a systematic review. If your program requires a formal search strategy (databases, search terms, PRISMA flow), add it here.

Instruments compared

Criterion	AHC-HRSN (CMS)	PRAPARE (NACHC)	Hunger Vital Sign
Length	10 core items (+ supplemental)	~21 items (core + optional)	2 items
Housing	✅ status + quality	✅ status + worry	—
Food	✅ (Hunger Vital Sign)	✅	✅
Transportation	✅	✅	—
Utilities	✅	✅	—
Interpersonal safety	✅ (HITS-based, scored)	✅ (optional)	—
Health literacy	—	—	—
ID/documents	—	—	—
Cost/licensing	Free (public, CMS)	Free (see NACHC terms)	Free
Origin setting	Clinical settings serving Medicare/Medicaid beneficiaries	Community health centers (FQHCs)	Pediatrics / general
Validation note	Items drawn from previously validated measures; TEP-reviewed (Billioux 2017)	Extensive field use; FHIR/ LOINC mappings	Sensitivity ~97 %, specificity ~83 % vs. USDA 18-item (Hager 2010)

Sources: SIREN comparison table; instrument documentation. Few social-risk tools have gone through full gold-standard psychometric development (SIREN). We found no validation study specific to unsheltered adults.

Key findings

1. **Brevity matters in mobile settings.** Both AHC-HRSN core and PRAPARE take about 5–10 minutes. AHC core covers all five of our required “core” domains in 10 items. PRAPARE adds demographics the clinic may already collect elsewhere.

2. **The housing item works differently for this population.** Nearly all patients will screen positive on "living situation." The useful information is the *type* of homelessness (unsheltered, shelter, doubled-up, vehicle), because that decides the referral (Coordinated Entry, shelter, hypothermia program). We therefore add a "where did you sleep last night" follow-up.
3. **Some items do not apply.** Housing-quality and utility-shutoff items assume a dwelling. For unsheltered patients they should be skipped, not scored as negative.
4. **Health literacy and ID/documents are not covered by any of the main tools.**
    - Health literacy: the **Single Item Literacy Screener (SILS)** (Morris 2006) is one item, published, and widely used.
    - ID/documents: we found no validated item. Lack of ID blocks access to shelter, benefits, Medicaid enrollment, and employment. We use a **locally developed** item and label it as unvalidated.
5. **Ask whether the patient wants help.** Screening positive does not mean the patient wants assistance. Studies find substantial mismatch between reported risk and interest in help (De Marchis 2020). Asking "would you like help with this?" respects autonomy and makes referral measures more meaningful.
6. **Screening is acceptable to most patients** when the reason is explained (De Marchis 2019, Byhoff 2019). Privacy and a clear purpose matter, especially for safety questions.
7. **Screening without a response pathway is the most common failure mode.** Navigation (in-person help) outperformed written information alone in an RCT (Gottlieb 2016). This supports a warm handoff plus follow-up, not just a handout.
8. **The federal measure landscape changed in 2025.** CMS removed the "Screening for Social Drivers of Health" quality measure (Quality ID 487) from the MSSP APP Plus set. NCQA's HEDIS **Social Need Screening and Intervention (SNS-E)** measure continues, with technical updates for MY 2026. This project's measures are internal QI measures modeled on these definitions, so they do not depend on federal reporting.

Recommendation

Use the **AHC-HRSN core 10 items** as the base, because they are free, public-domain, short, and cover the required domains. Adapt them for homelessness:

Domain	Item source	Adaptation
Housing status	AHC Q1	+ "where did you sleep last night" (local) for referral routing
Housing quality	AHC Q2	Skip if no steady place to live
Food	AHC Q3–4 (Hunger Vital Sign)	None
Transportation	AHC Q5	None
Utilities	AHC Q6	Skip if no steady place to live
Interpersonal safety	AHC Q7–10 (HITS)	Ask only in private; otherwise defer and document
Health literacy	SILS (Morris 2006)	None
ID/documents	Local item	Flag as unvalidated; review with patient advisor
Desire for help	Local item (modeled on De Marchis 2020)	Asked once per positive domain

Alternative: if the clinic is an FQHC or Health Care for the Homeless (HCH) grantee already reporting PRAPARE (UDS), use PRAPARE instead so the data stay consistent. **Confirm this before go-live** (see 09).

Gaps to state in the write-up

- We found no instrument validated specifically among unsheltered adults.
- The ID/documents item is unvalidated.
- With n≈30, findings about need prevalence are descriptive only.

References

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> Verify each citation against the full text before academic submission. Some details here (for example, page ranges) come from secondary sources.

02 — Standardized Social Needs Screening Questionnaire

Version: 0.1 (draft for patient-advisor and clinic review) **Administration:** staff-read aloud (default) or self-completed. About 5–7 minutes. **Eligible:** adults ≥ 18 at a clinic visit who have not completed a screen in the past 90 days (see 03).

Item text for Q1–Q10 is from the CMS **Accountable Health Communities HRSN Screening Tool** (public). Q11 is the **Single Item Literacy Screener** (Morris 2006). Items marked **(local)** are developed for this project and have not been validated.

A printable version is in `templates/screening-form.md`.

Intro script (read aloud)

> "We ask everyone we see the same few questions about things like food, transportation, and safety, because these affect health. Your answers are private and won't affect the care you get today. You can skip any question. If something comes up, we can connect you with help, only if you want it."

Housing

Q1. What is your living situation today? *(AHC)* - ☐ I have a steady place to live - ☐ I have a place to live today, but I am worried about losing it in the future → **POSITIVE (at risk)** - ☐ I do not have a steady place to live (I am temporarily staying with others, in a hotel, in a shelter, living outside on the street, on a beach, in a car, abandoned building, bus or train station, or in a park) → **POSITIVE (unstable)**

Q1a. Where did you sleep last night? *(local, used for referral routing, not scoring)* ☐ Outside / street / park / woods ☐ Vehicle ☐ Emergency shelter / hypothermia site ☐ Staying with friends/family ☐ Motel/hotel ☐ Own apartment/room ☐ Other: ____

Q2. Think about the place you live. Do you have problems with any of the following? *(AHC, **skip if Q1 = "do not have a steady place to live")*** ☐ Pests ☐ Mold ☐ Lead paint or pipes ☐ Lack of heat ☐ Oven or stove not working ☐ Smoke detectors missing or not working ☐ Water leaks ☐ None of the above → **POSITIVE** if any problem is selected.

Food

Q3. Within the past 12 months, you worried that your food would run out before you got money to buy more. *(AHC / Hunger Vital Sign)* ☐ Often true ☐ Sometimes true ☐ Never true

Q4. Within the past 12 months, the food you bought just didn't last and you didn't have money to get more. *(AHC / Hunger Vital Sign)* ☐ Often true ☐ Sometimes true ☐ Never true

→ **POSITIVE** if "Often true" or "Sometimes true" on **either** Q3 or Q4.

Transportation

Q5. In the past 12 months, has lack of reliable transportation kept you from medical appointments, meetings, work, or from getting things needed for daily living? *(AHC)* ☐ Yes → **POSITIVE** ☐ No

Utilities

Q6. In the past 12 months has the electric, gas, oil, or water company threatened to shut off services in your home? *(AHC, **skip if Q1 = "do not have a steady place to live")*** ☐ Yes → **POSITIVE** ☐ No ☐ Already shut off → **POSITIVE**

Interpersonal safety

> **Ask only if the patient is alone and the conversation cannot be overheard.** If privacy is not possible, mark "Deferred: no privacy" and retry at the next visit.

(AHC, adapted from HITS) How often does anyone, including family and friends:

	Never (1)	Rarely (2)	Sometimes (3)	Fairly often (4)	Frequently (5)
Q7. physically hurt you?	☐	☐	☐	☐	☐
Q8. insult or talk down to you?	☐	☐	☐	☐	☐
Q9. threaten you with harm?	☐	☐	☐	☐	☐
Q10. scream or curse at you?	☐	☐	☐	☐	☐

→ **POSITIVE** if Q7–Q10 total **≥ 11**. Follow the safety pathway in 03 §6 and do not wait for the full score if imminent danger is disclosed.

Health literacy

Q11. How often do you need to have someone help you when you read instructions, pamphlets, or other written material from your doctor or pharmacy? *(SILS)* ☐ Never ☐ Rarely ☐ Sometimes ☐ Often ☐ Always → **POSITIVE** if "Sometimes", "Often", or "Always". *A positive result changes how we communicate (teach-back, verbal instructions, pictograms). It is not a referral.*

ID / documents *(local, unvalidated)*

Q12. Do you have these documents with you or somewhere safe that you can get to?

Document	Have it	Don't have it	Want help getting it?
Photo ID (state ID / driver's license)	☐	☐	☐ Yes ☐ No
Social Security card	☐	☐	☐ Yes ☐ No
Birth certificate	☐	☐	☐ Yes ☐ No
Health insurance / Medicaid card	☐	☐	☐ Yes ☐ No

→ **POSITIVE** if any document is missing.

Desire for help *(local)*

Q13. For each area that came up, would you like help with it today? ☐ Housing ☐ Food ☐ Transportation ☐ Utilities ☐ Safety ☐ ID/documents ☐ None right now

Contact for follow-up *(local, operational)*

Q14. What is the best way for us to reach you in the next few weeks? ☐ Phone/text: _____ ☐ Through shelter/case manager: _____ ☐ At the next clinic stop ☐ Other: _____ ☐ Prefer no follow-up

---

Scoring summary

Domain	Positive if	Not applicable when
Housing (status)	Q1 ≠ steady	—
Housing (quality)	Any Q2 problem	Q1 = no steady place
Food	Q3 or Q4 = often/sometimes	—
Transportation	Q5 = Yes	—
Utilities	Q6 = Yes / already shut off	Q1 = no steady place
Safety	Q7–10 sum ≥ 11	Deferred (no privacy) → record `deferred`
Health literacy	Q11 ≥ sometimes	—
ID/documents	Any document missing	—

Screen "complete" means Q1, Q3–Q5, Q11, and Q12 are answered (or patient declined that item), and safety is either answered or recorded as deferred. Skipped-by-logic items count as complete.

Coding for data entry

See data/data-dictionary.md. Optional standard codes for an EHR (Phase 2): LOINC panel 96777-8 (AHC HRSN screening tool) and ICD-10-CM Z55–Z65 for documented needs (e.g., Z59.0x homelessness, Z59.41 food insecurity, Z59.82 transportation insecurity). **Verify the current code set before use.**

Translation

The AHC tool is available in multiple languages from CMS. Use professionally translated versions where available. The local items need translation and back-translation, especially into Spanish. Fairfax County's Coordinated Services Planning line offers many languages for referrals (04).

03 — Screening Protocol (Standard Operating Procedure)

Purpose: Every eligible patient is screened the same way, and every positive screen gets the same response.

1. Eligibility

Rule	Definition
Logged visit	Every visit by a patient ≥ 18 for any clinical encounter. Log **all** adult visits, not only screened ones, because they form the denominator for P1.
Screen **due**	No complete screen in the prior **90 days**
Screen not due	Record `screen_status = not_due`. The visit still counts toward P1 (up to date).
Excluded from denominator	Patient medically unstable / transferred to ED; patient unable to participate (e.g., acute intoxication, severe distress). *Document the reason and retry at the next visit.*
Patient declines	**Stays in the denominator** (counts as not screened) but is recorded as `declined`. Decline rate is a balancing measure.

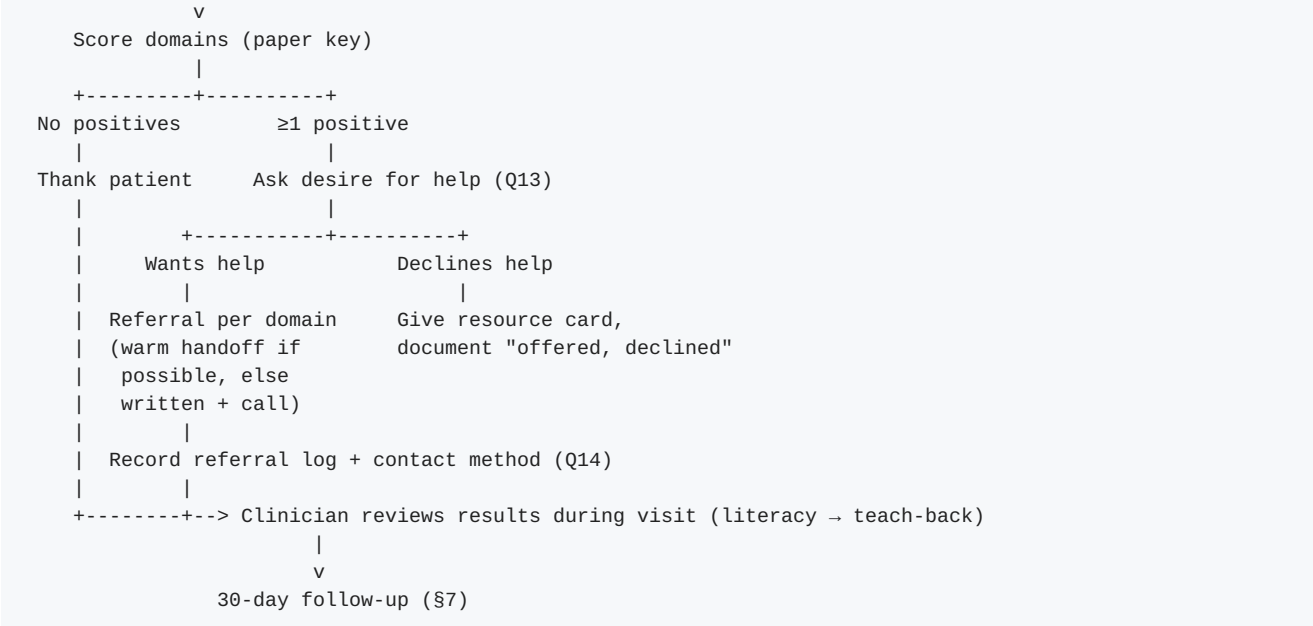
Why 90 days: living situation and needs change quickly in this population. Annual rescreening (common in primary care) would miss changes. Adjust the interval if the clinic's visit frequency differs.

2. Roles

Step	Primary	Backup
Check whether screen is due	Intake/registration staff	Screening champion
Administer screen	MA / RN / CHW (trained)	Clinician
Review positives + ask desire for help	Person who screened	Clinician
Warm handoff / referral	Referral owner (CHW / case manager)	Screener
Safety-positive response	Clinician (licensed)	Clinic lead
30-day follow-up	Referral owner	Screening champion
Data entry (logs)	Screener (same day)	Project lead (weekly audit)

3. Workflow

```
Arrival --> Intake: screen due? (log / flag)
               | no ------------------------------> usual visit
               | yes
               v
         Private space available? (needed for safety items Q7-Q10; if not, mark deferred)
               |
               v
     Administer Q1–Q14 (read aloud)  -- patient declines -> record "declined", offer resource card
               |
```



4. Administration standards

- Read questions **exactly as written**, without paraphrasing the scored items. A brief clarification is OK after the scripted question.
- Use a professional interpreter (phone/video) for non-English speakers. Do not use family or friends as interpreters for safety items.
- Use the intro script from 02.
- Record answers on the paper form, then transcribe to the screening log the same day.

5. Responding to positive screens (non-safety)

Domain	First-line response	Referral target (see 04)
Housing: unsheltered / shelter / vehicle	Warm call to shelter intake / Coordinated Entry; hypothermia program Dec–Mar	Fairfax OPEH / shelters / Hypothermia Prevention Program
Housing: at risk of losing housing	Eviction-prevention referral	Coordinated Services Planning (CSP)
Housing quality	Referral to CSP	CSP
Food	Same-day food if available; pantry referral; SNAP application help	WFCM; DFS (SNAP)
Transportation	NEMT info if Medicaid; reduced-fare programs	Cardinal Care NEMT; TOPS / TRIP
Utilities	Energy assistance referral	DFS Energy Assistance; CSP
Health literacy	**Clinical response:** teach-back, verbal instructions, pictograms, plain-language handouts	(no external referral)
ID/documents	Help with application; fee-waiver eligibility	VA DMV / Vital Records (free if < 21 and homeless); CSP

Referral types (record one): `warm_handoff` (staff calls or introduces the patient during the visit), `direct_referral` (staff sends the referral), `information_only` (written resource given).

6. Safety pathway (HITS ≥ 11, or any disclosure of current danger)

1. Thank the patient for sharing. Make sure the patient is in a private space.
2. Clinician assesses **immediate danger**. If the patient is in imminent danger, call **911**.
3. Offer the **Fairfax County Domestic and Sexual Violence 24-Hour Hotline: 703-360-7273 (TTY 711)**. Offer to call with the patient.
4. Do **not** give written materials that could put the patient at risk if found. Ask first.
5. **Mandatory reporting:** Virginia requires certain professionals to report suspected abuse, neglect, or exploitation of adults aged 60+ or incapacitated adults to Adult Protective Services (Va. Code § 63.2-1606). **Confirm reporting obligations with the clinic's compliance lead before go-live** (see 08).
6. Document per clinic policy in the clinical record. Record only `safety_positive = Y` and the referral in the QI log, with no narrative detail.

7. Follow-up (closed loop)

- At **14 ± 7 days**, and again by **30 days** if needed, the referral owner contacts the patient using the Q14 method (phone, shelter or case manager, or the next clinic stop).
- Ask: "Were you able to connect with ___? Did you get what you needed?"
- Record the `followup_status`: `connected`, `not_connected`, `unable_to_reach`, `declined_followup`, or `pending`.
- If the patient was not connected, re-refer or remove the barrier, for example with a bus fare or a call made together.

8. Rescreening

- Rescreen every 90 days, or sooner if the patient reports a major change (eviction, new shelter, safety event).

9. Documentation

What	Where (paper-first)	Phase 2
Completed form	Locked binder / scanned to chart	EHR flowsheet
Screening log	`screening_log.csv` (coded with study ID)	Export
Referral log	`referral_log.csv`	Closed-loop referral platform
Study ID ↔ patient key	**Separate, locked**, held by clinic only	—

04 — Resource Directory: Chantilly / Fairfax County, VA

> **Last verified:** 2026-11-05 (online sources, listed per entry). Phone numbers, hours, and eligibility change. **Call each resource to confirm before go-live, then re-verify quarterly.** Record the verification date in the right column.
>
> Fairfax County's **Coordinated Services Planning (CSP) line, 703-222-0880**, is the single front door for most county human services and can help when you don't know where to start.

Emergency / general

Resource	What	Contact	Notes	Source	Verified by clinic
Emergency	Imminent danger, medical emergency	**911**			
Coordinated Services Planning (CSP)	Front door for food, shelter, rent, financial, healthcare referrals	**703-222-0880**, TTY 711; M–F 8:00–4:30	Multilingual specialists (Spanish, Amharic, Farsi/Dari, Korean, Urdu, and more)	fairfaxcounty.gov/neighborhood-community-services/coordinated-services-planning	☐
Fairfax County non-emergency	Report an unsheltered person at risk of hypothermia	703-691-2131, TTY 711		VirginiaNavigator listing for OPEH	☐

Housing / shelter

Resource	What	Contact	Notes	Source	Verified
Office to Prevent and End Homelessness (OPEH)	County homelessness office	703-246-5000, TTY 711; M–F 9–5		fairfaxcounty.gov/homeless	☐
Embry Rucker Community Shelter (Cornerstones)	Shelter, closest to the western/northern county; hypothermia site	11975 Bowman Towne Dr, Reston · **703-437-1975** · 24/7	Single adults may self-refer	Fairfax Hypothermia flyer 2025–26	☐
Bailey's Shelter & Supportive Housing	Single-adult shelter	5914 Seminary Rd, Falls Church · 703-820-7621 · 24/7		Fairfax Hypothermia flyer 2025–26	☐
Eleanor U. Kennedy Shelter (New Hope Housing)	Single-adult shelter	9155 Richmond Hwy, Alexandria · 703-799-0200		VirginiaNavigator	☐

Resource	What	Contact	Notes	Source	Verified
Hypothermia Prevention Program	Winter shelter for any adult, **Dec 1–Mar 31**	Fairfax/Centreville rotating faith sites: 703-352-5090 (ext. 303 Spanish) per VirginiaNavigator; Reston: North County Human Services Center, 1850 Cameron Glen Dr, Rm 200, 5 pm–7 am	Text **FFXOPEH to 888-777** for overflow/heat/cold alerts	fairfaxcounty.gov/homeless	☐
Families with children	Shelter eligibility + referral	CSP 703-222-0880		fairfaxcounty.gov/homeless	☐
Eviction prevention / rent help	Emergency rental assistance	CSP 703-222-0880		CSP page	☐

Food

Resource	What	Contact	Notes	Source	Verified
Western Fairfax Christian Ministries (WFCM) — Chantilly Free Food Market	Food pantry for western Fairfax (Chantilly, Centreville, Clifton, Oak Hill, Herndon, Fairfax, Fairfax Station)	4511 Daly Dr, Suite J, Chantilly 20151 · **703-988-9656**	Hours vary; confirm current hours and eligibility documents	wfcmva.org	☐
WFCM — Centreville Free Food Market	Food pantry	Centreville (see wfcmva.org)		wfcmva.org	☐
SNAP (DFS)	Benefits application	DFS 703-324-7500, TTY 711; M–F 8–4:30; or commonhelp.virginia.gov	A shelter or mailing address can be used	fairfaxcounty.gov/familyservices	☐

Transportation

Resource	What	Contact	Notes	Source	Verified
Medicaid (Cardinal Care) NEMT	Free rides to covered medical appointments	Number depends on the patient's health plan (on member card). Fee-for-service: **1-866-386-8331** (ModivCare)	Book ≥ 5 days ahead; bus tickets possible	DMAS / VirginiaNavigator	☐

Resource	What	Contact	Notes	Source	Verified
TOPS (Transportation Options, Programs & Services)	Discounted transportation debit card for eligible low-income, older, or disabled residents	**703-222-9764**, TTY 711; M–F 8–4:30		fairfaxcounty.gov NCS	☐
TRIP (Transit Ridership Incentive Program)	50 % off Fairfax Connector; free transfers to/from Metrobus and Metrorail	Apply via Fairfax Connector / TOPS	Eligibility criteria apply (income, age, SSI, MetroAccess); check current criteria	fairfaxcounty.gov NCS / news	☐

Utilities

Resource	What	Contact	Notes	Source	Verified
DFS Energy Assistance	Heating/cooling help, energy crisis	703-324-7500, TTY 711	Seasonal application windows	fairfaxcounty.gov/familyservices	☐
CSP	Emergency financial help	703-222-0880			☐
WFCM	Some utility assistance reported	703-988-9656	Confirm availability	wfcmva.org news	☐

Interpersonal safety

Resource	What	Contact	Notes	Source	Verified
Fairfax County Domestic & Sexual Violence 24-Hour Hotline	Crisis line, shelter, advocacy	**703-360-7273**, TTY 711	24/7	fairfaxcounty.gov/homeless	☐
Adult Protective Services	Required reports for adults 60+ / incapacitated adults	Via Fairfax DFS (confirm number)	See 08		☐

ID / documents

Resource	What	Contact	Notes	Source	Verified
Virginia DMV — No-cost credentials for homeless youth	Free ID card, driver's license, and Virginia birth certificate	dmv.virginia.gov/licenses-ids/no-cost-credentials	**Under 21**, homeless, VA resident; needs a *Virginia Affidavit Certifying Homeless Youth Status*. **The clinic may be able to certify it** if it is a publicly funded or CoC-recognized agency. Effective July 1, 2025.	VA DMV	☐
Virginia DMV — Standard ID card (adults 21+)	State ID	dmv.virginia.gov	Fee applies (reported ~$10, confirm)	Secondary source	☐
Fairfax County Vital Records (Health Dept.)	Certified VA birth certificates	10777 Main St, Fairfax · 703-246-2411	Fee applies for adults (confirm)	fairfaxcounty.gov/health/vital-records	☐
Social Security Administration	Replacement SS card (no fee)	ssa.gov / local office	Needs identity proof	ssa.gov	☐
Medicaid card / enrollment	Apply or replace card	Cover Virginia / commonhelp.virginia.gov; DFS 703-324-7500			☐
CSP	Help with document fees from local funds, if available	703-222-0880	Ask		☐

Gaps to resolve before go-live

- ☐ Confirm whether the clinic qualifies to certify the youth homeless affidavit.
- ☐ Identify a local source of funds for ID fees for adults 21+, for example a faith-community partner.
- ☐ Confirm the clinic's local Coordinated Entry process for adults: shelter self-referral vs. CSP.
- ☐ Add a phone/cell-service resource (federal Lifeline program) so patients can be reached for follow-up.
- ☐ Print a pocket-sized **resource card** (English/Spanish) for patients.

05 — Staff Training Plan

Learning objectives

After training, staff will be able to: 1. Explain why every patient is screened the same way (equity and reliability). 2. Determine whether a screen is due, using the 90-day rule. 3. Administer Q1–Q14 verbatim, with trauma-informed language and correct skip logic. 4. Score each domain correctly with the scoring key. 5. Ask about desire for help and make the right referral type. 6. Follow the safety pathway, including privacy and reporting obligations. 7. Complete the screening and referral logs accurately.

Format

Session	Length	Audience	Method
1. Why and what	30 min	All staff and volunteers	Short talk + baseline data + patient-advisor perspective
2. Hands-on administration	45 min	Anyone who screens	Demo → paired role-play with 3 scripted cases → feedback
3. Safety pathway	20 min	Screeners + clinicians	Case discussion; reporting rules from the compliance lead
4. Logs and follow-up	20 min	Screeners + referral owner	Practice filling in the logs from sample forms
Huddle refreshers	5 min	All	At the start of each clinic session during PDSA cycles

Total: about 2 hours. It can run as one block or be split across clinic days. Add it to new-staff and volunteer onboarding for sustainment.

Trauma-informed communication (key points)

- Explain why you are asking, and that the patient can skip any question.
- Use neutral, non-judgmental phrasing, for example “Where did you sleep last night?” rather than “Are you homeless?”
- Do not promise outcomes you cannot deliver. Say “I can connect you with…” instead of “They will give you…”.
- Respect a “no”. Offer the resource card and let the patient know they can ask at any visit.
- Make sure the space is private before asking safety questions.

Role-play cases

1. **Ms. A:** 54 years old, sleeping in her car, food insecure, missed two appointments for lack of a ride, and has Medicaid. → Housing routing (vehicle), food, NEMT.
2. **Mr. B:** 19 years old, staying with friends, lost his wallet with his ID and Social Security card. → Free youth credentials (< 21), SS card replacement, housing at-risk.

3. **Mx. C:** in a shelter. During the safety items, a partner is waiting nearby. → Defer the safety items, document “deferred: no privacy”, and set up a private retry.

Competency check (observed, per screener)

#	Item	Met?
1	Uses intro script	☐
2	Reads scored items verbatim	☐
3	Applies skip logic (Q2, Q6) correctly	☐
4	Ensures privacy before Q7–Q10	☐
5	Scores all domains correctly	☐
6	Asks desire-for-help for each positive domain	☐
7	Chooses appropriate resource and referral type	☐
8	Records contact method (Q14)	☐
9	Completes logs without errors	☐

Pass = 9/9 on one observed real or simulated screen. Re-observe if fewer than 9.

Training evaluation

- Pre/post knowledge quiz (8–10 items covering eligibility, scoring, and the safety pathway).
- Confidence rating (1–5) for “I feel confident screening for social needs”, before and after.
- % of screeners who pass the competency check before go-live (target 100 %).

Materials checklist

- ☐ Laminated job aid (workflow + scoring key), see 03
- ☐ Printed forms (templates/screening-form.md)
- ☐ Resource directory and patient resource cards (04)
- ☐ Interpreter-line instructions
- ☐ Knowledge quiz and competency checklist

06 — Implementation Plan

Framework: **Model for Improvement** (Langley et al.) with sequential PDSA cycles, reported with **SQUIRE 2.0**.

Three questions

1. **What are we trying to accomplish?** See the SMART aim in 00.
2. **How will we know a change is an improvement?** See the measures in 07.
3. **What changes can we make?** See the driver diagram below.

Driver diagram

Aim	Primary driver	Change ideas
≥ 80 % of visits with an up-to-date screen; **≥ 90 %** of positive screens offered a referral	**Standard tool & workflow**	Adopt the adapted AHC-HRSN form · "Screen due?" flag at intake · Scoring key on the form
	Staff capability & buy-in	2-hr training + competency check · Screening champion · Huddle reminders + weekly run chart
	Patient trust & privacy	Intro script, opt-out · Private space for safety items · Patient-advisor review of wording
	Reliable response to needs	Verified resource directory · Warm handoff when possible · Resource card (EN/ES) · 30-day follow-up tracking

Phases (default 16 weeks; timeline not yet decided)

Phase	Weeks	Key activities	Exit criteria
0. Prepare	−4 to 0	Sponsor approval; QI/IRB determination; verify resources; patient-advisor review; train staff; print materials	Approvals done; 100 % screeners pass competency
1. Baseline	1–4	Measure current practice: chart audit of all adult visits (was *any* social need asked and documented? referral?)	≥ 4 weekly data points
2. PDSA 1	5–8	Champion screens every patient who is due, at intake (paper)	Screening ≥ 60 %; workflow time known
3. PDSA 2	9–12	Spread to all trained staff; add desire-for-help and referral-offer step	Screening ≥ 80 %; referral offer ≥ 90 %
4. PDSA 3	13–16	Add 14-/30-day follow-up and closed-loop tracking	Follow-up documented ≥ 50 %

Phase	Weeks	Key activities	Exit criteria
5. Sustain	17+	Add to onboarding; monthly audit; hand off to clinic owner	Sustained ≥ 80 % for 3 months

PDSA cycle plans

PDSA 1: Champion-led screening at intake

- **Plan:** A single trained champion screens every adult who is due, at intake with the paper form. *Prediction:* screening rises from about 25 % to at least 60 %, with fewer than 8 minutes added per visit.
- **Do:** Run for 4 weeks. Log visit time, declines, and barriers (no private space, interpreter needed).
- **Study:** Review the weekly run chart, decline rate, and time per screen. Debrief the champion.
- **Act:** Adopt, adapt (for example, move screening to the waiting period), or abandon.

PDSA 2: Spread + referral offer

- **Plan:** All trained staff screen. Each positive domain gets the desire-for-help question and a referral offer. *Prediction:* screening ≥ 80 % and referral offer ≥ 90 %.
- **Study:** Compare screening rate by screener role (fidelity) and the referral funnel.

PDSA 3: Closed-loop follow-up

- **Plan:** The referral owner follows up at 14 ± 7 days and again by 30 days. *Prediction:* ≥ 50 % of accepted referrals have a documented outcome.
- **Study:** Track the unable-to-reach rate and which contact methods work.

Use templates/pdsa-worksheet.md for each cycle.

Stakeholder and communication plan

Who	What	When
Clinic staff	Run chart on the wall/van + huddle	Weekly
Clinic sponsor	1-page update	Every 4 weeks
Faculty advisor	Progress + data	Per program schedule
Community partners (WFCM, shelters, CSP)	Heads-up on expected referral volume; feedback	Before go-live + monthly
Patients	Resource card; “we ask everyone” signage	Ongoing

Barriers and mitigations

Barrier	Mitigation
Limited private space in the van	Use an outdoor/curtained area for safety items, or defer and record “deferred”
Interpreters	Phone interpreter line; translated forms
Volunteer/staff turnover	Job aid; champion; onboarding module

Barrier	Mitigation
Resources at capacity	Re-verify quarterly; track “not connected” reasons
Patients without phones	Collect a shelter/case-manager contact; follow up at the next stop; Lifeline referral

Sustainability

- Add the screening to the standard visit template.
- Assign an owner for monthly audit (10 random charts).
- Re-verify the resource directory quarterly.
- Phase 2 tool decision (09).

07 — Measurement Plan

Unit of analysis: weekly, for run charts. **Data sources:** `screening_log.csv`, `referral_log.csv` (data dictionary). **Analysis:** `analysis/analyze.py`.

Measure set

ID	Type	Measure	Numerator	Denominator	Target
P1	Process (primary)	Up-to-date screening rate	Adult visits where the patient has a **complete** screen at that visit **or** within the prior 90 days	All adult visits (incl. declines; excl. documented medical exclusions)	≥ 80 %
P1a	Process	Due-screen completion	Visits where a screen was **due** and was completed at that visit	Visits where a screen was due (no complete screen in prior 90 days)	≥ 80 % (pre/post; small n)
P1b	Process	Panel coverage	Patients with ≥ 1 complete screen to date (cumulative)	Patients seen to date	Monitor
P2	Process	Referral offer rate	Positive-screen visits where a referral was offered for **every** actionable positive domain*	Visits with ≥ 1 actionable positive domain	≥ 90 %
P3	Process	Referral acceptance	Referrals accepted	Referrals offered	Monitor (no target; autonomy)
P4	Process	Fidelity: safety items	Complete screens with safety answered privately	Complete screens	Monitor; reduce `deferred`
O1	Outcome	Follow-up documented	Accepted referrals with a follow-up status other than `pending` by day 30	Accepted referrals with ≥ 30 days elapsed	≥ 50 %
O2	Outcome	Connection rate	Accepted referrals with `connected` by day 30	Accepted referrals with ≥ 30 days elapsed	Monitor (establish baseline)
B1	Balancing	Decline rate	Due visits where the patient declined	Visits where a screen was due	Monitor; investigate if > 20 %
B2	Balancing	Added visit time	Median screening minutes	Screens with time recorded	≤ 8 min
B3	Balancing	Staff burden	Mean rating on “screening is manageable” (1–5)	Staff surveyed (start/end)	≥ 3.5

ID	Type	Measure	Numerator	Denominator	Target
D1	Descriptive	Need prevalence by domain	Patients positive per domain	Patients with ≥ 1 complete screen	—

*Health literacy is **not** an actionable referral domain. Its response is clinical (teach-back), so it is excluded from P2.

Why P1 is “up to date” rather than “screened at this visit”

With a panel of about 30 patients and a 90-day rescreening interval, most patients are *not due* at most visits once screening starts. A “screened this visit” rate would have weekly denominators of 0–3, too small for a run chart. P1 asks the question that matters for every visit: *does the care team know this patient’s current social needs?* It keeps weekly denominators at the visit volume. P1a (did we screen when due?) is reported pre/post or biweekly. If the clinic actually sees about 30 **different** patients per session, P1a becomes plottable weekly. Confirm the panel size and visit volume.

Baseline data collection

Baseline uses a chart audit of every adult visit during the baseline weeks. Count a visit as “screened” only if **all** core domains (housing, food, transportation, utilities or N/A, safety or deferred, health literacy, ID/documents) are documented. Partial screening is recorded in `notes` but counts as not complete. This matches the problem statement that questions vary by provider. Also record whether any single domain was asked, to describe how inconsistent the baseline is.

Run chart rules (Perla, Provost & Murray 2011)

Plot the weekly percentage, with the centerline at the **baseline median**. A non-random signal is: 1. **Shift:** 6 or more consecutive points all above or all below the median. Points on the median don’t count and don’t break the run. 2. **Trend:** 5 or more consecutive points all going up or all going down. Repeated values don’t count. 3. **Too few or too many runs:** compare against the Swed–Eisenhart table for the number of useful points. 4. **Astronomical point:** an obviously extreme value, judged visually.

Annotate each chart with the PDSA start dates. After a sustained shift, the median may be re-baselined.

Why run charts and not p-values: with about 30 patients and 10–20 visits a week, the data are too small and too autocorrelated for meaningful hypothesis tests. Run charts detect improvement over time, which is the standard approach for QI projects. Pre/post percentages are reported descriptively, with 95 % Wilson confidence intervals.

Small weekly denominators: if a week has fewer than 5 eligible visits, combine it with the next week or plot biweekly. The analysis script flags these weeks.

Data quality

- Weekly: the project lead checks 100 % of logs against forms during PDSA 1, then a 20 % sample.
- Missing data: a screen with a missing core item counts as **incomplete**. Report the missing-data rate.
- The analysis script validates the schema and allowed values, and lists errors before computing measures.

Stratification (if n allows)

Report by screener role, language, and sleeping location. This checks whether screening is consistent across groups, which is the core equity goal. With small numbers, report counts and avoid percentages for cells with n < 5.

Reporting

- Weekly: run chart P1 (+ P2 after PDSA 2) posted for staff.
- End of project: all measures in a SQUIRE 2.0 report. The final run charts come from `analysis/analyze.py`.

References

- Perla RJ, Provost LP, Murray SK. The run chart: a simple analytical tool for learning from variation in healthcare processes. *BMJ Qual Saf.* 2011;20(1):46-51.
- Langley GJ, Moen RD, Nolan KM, et al. *The Improvement Guide.* 2nd ed. Jossey-Bass; 2009.
- Ogrinc G, Davies L, Goodman D, et al. SQUIRE 2.0: revised publication guidelines from a detailed consensus process. *BMJ Qual Saf.* 2016;25(12):986-992.

08 — Ethics, Privacy, and QI Determination

> This document is guidance for planning. It is **not legal advice**. Confirm every item with the faculty advisor, the institution's IRB, and the clinic's compliance or privacy officer.

1. QI vs. research determination

Academic QI projects usually need a formal **determination** from the student's institution, either "QI, not human-subjects research" or IRB review. Get it **before** collecting any data, including baseline chart audits.

Features of this project that point to QI: - The aim is to improve local practice using an established standard (AHC-HRSN). - Everyone eligible gets the improved process. There is no randomization or withholding. - Data are used for local improvement.

Features that may call for IRB review: - An intent to publish generalizable findings. Many IRBs still allow QI publication, so ask. - Collecting data beyond routine care, such as patient interviews or surveys for research. - The vulnerable population. People experiencing homelessness may be treated as needing added protections.

Action: submit the institution's QI determination form with this charter, protocol, and measurement plan.

2. Privacy (HIPAA)

- The clinic is likely a HIPAA covered entity. The student needs a role in the clinic's workforce (for example, a student or volunteer agreement) or a data use arrangement to access PHI.
- **Minimum necessary:** the QI logs use a **study ID** only, with no names, DOB, addresses, or phone numbers. The study-ID-to-patient key stays at the clinic, locked, and separate from the logs.
- **The coded logs are still PHI.** Replacing names with study IDs does not de-identify them. `visit_date`, `referral_date`, `followup_date`, and the `week` number derived from them are date elements. HIPAA's Safe Harbor method requires removing all date elements except the year (45 CFR 164.514(b)). Converting dates to week numbers does **not** meet that standard. Coded logs with dates are at most a **Limited Data Set** (45 CFR 164.514(e)).
- Phone numbers from Q14 stay on the paper form and in the clinic record. **Never put them in the CSV logs or this repository.**
- **Never commit real patient data to GitHub.** This repo is for templates and synthetic data only. `.gitignore` blocks `data/real/`.
- Small cells: with about 30 patients, suppress or merge cells with n < 5 in reports to reduce re-identification risk.

Getting data out of the clinic

Before the baseline audit, choose one route in writing with the clinic's privacy officer:

Route	What leaves the clinic	Requirement
1. Analyze on site	Aggregate results only	Student has workforce status (student or volunteer agreement, HIPAA training) and works only on clinic-controlled storage
2. Clinic runs the analysis	Aggregate results only	Clinic staff run `analysis/analyze.py` and share the summary
3. Limited Data Set	Coded logs, including dates	Signed **Data Use Agreement** between the clinic and the student or school

Aggregate results (run charts, percentages, the summary) go into the thesis or report only after the privacy officer reviews them. `analyze.py` suppresses small cells in the prevalence table, but the weekly counts in `weekly_measures.csv` and the run charts are **not** suppressed. If weekly counts are small, report percentages or monthly totals. See 10 for the full plan.

3. Consent and autonomy

- Screening is part of routine care, so a separate research consent is normally not required for QI. The patient may still decline any or all items without any effect on care.
- Ask about desire for help separately from risk. Patients may decline referrals.
- Do not give written materials on safety resources without asking whether it is safe for the patient to carry them.

4. Safety disclosures and mandated reporting (Virginia)

- **Adults 60+ or incapacitated adults:** Virginia law makes certain professionals mandated reporters of suspected abuse, neglect, or exploitation to Adult Protective Services (Va. Code § 63.2-1606). Confirm which clinic roles this applies to.
- **Other adults (IPV):** confirm with compliance whether any injury-reporting rules apply to the clinicians. Otherwise, follow a patient-centered response: hotline, safety planning, and patient choice.
- **Minors** are out of scope (adults only). If a minor is encountered, follow child protective services reporting rules.
- Document the clinic's final policy in 03 §6 before go-live.

5. Equity considerations

- Screen everyone the same way. That consistency is the purpose of the project.
- Provide language access (interpreters and translated forms).
- Avoid documentation that could be used against patients. Keep QI logs minimal and coded.
- Include a patient or lived-experience advisor in reviewing wording and acceptability.

6. Checklist

- [ ] Institutional QI/IRB determination letter on file
- [ ] Clinic data-access agreement / workforce status for student
- [ ] Route for getting data out of the clinic chosen in writing (on-site analysis, clinic-run analysis, or Limited Data Set + DUA)

- [ ] Compliance sign-off on safety pathway and reporting
- [ ] Data storage location approved (encrypted, clinic-controlled)
- [ ] Data retention/destruction plan for the linking key

09 — Phase 2 Decision Guide: What Should We Build?

Phase 1 is paper-first so the project can start without depending on any system. Phase 2 decides whether a tool would improve reliability, and which one.

Open questions (must answer first)

#	Question	Why it matters	Answer
1	**What does the clinic use to document today?** Paper, EHR (which one?), spreadsheet?	Decides whether we build inside an EHR or standalone	*Unknown — find out*
2	Does the clinic have tablets/ phones and reliable cellular data on the van?	Offline capability may be required	
3	Is the clinic an FQHC / Health Care for the Homeless grantee reporting UDS?	May require PRAPARE instead of AHC-HRSN	
4	Does the county or region use a closed-loop referral platform (e.g., Unite Us, findhelp)?	Integrate with it rather than build	
5	Who will own and maintain any tool after the student leaves?	Avoid building something unsupported	
6	What is the project timeline?	Limits how much can be built and tested	*Not decided*
7	Where may data be stored (HIPAA-compliant, clinic-controlled)?	Rules out many free tools	
8	Will patients self-administer on a tablet, or will staff read the questions?	UX design, literacy	
9	Is there a confirmed clinic partner and an approved route for getting data out of the clinic?	No real data or testing without one; see 10	*Not yet*

Build options

Option	Effort	Pros	Cons	Fits when
A. Stay on paper + spreadsheet (this repo's templates + `analyze.py`)	Very low	Zero cost; works offline; no IT approval	Transcription errors; manual follow-up tracking	No devices or IT support; short timeline
B. EHR-native form/ flowsheet	Low–medium (EHR analyst)	Data in the chart; reportable; sustainable	Needs vendor/IT support; less flexible	Clinic has an EHR with a configurable template

Option	Effort	Pros	Cons	Fits when
C. REDCap or similar secure survey tool	Low–medium	HIPAA-capable via institution; offline mobile app; built-in exports	Separate from the chart; institution must host	Academic partner with REDCap; no EHR flexibility
D. Existing referral platform	Low (config)	Closed-loop referrals built in	Licensing cost; partner adoption varies	Region already uses one
E. Custom app (offline PWA: screening + referral log + dashboard)	High	Tailored workflow and dashboard	Security, hosting, HIPAA, and maintenance burden	Only if A–D are not feasible and there is a long-term owner

Current recommendation: start with **A**. Move to **C** (REDCap) or **B** (EHR) when Question 1 is answered. Option E is the last resort. If a custom tool is built, a **QI dashboard** that reads the coded CSV logs (no direct identifiers) and runs on clinic-controlled storage is the safest and most useful thing to build first.

What Phase 1 already provides

- A data schema (data dictionary) that every option can export to.
- Analysis code that produces the run charts from any option's CSV export.
- A paper form that can be converted directly into REDCap or EHR fields.

10 — Real Data Plan: Replacing the Synthetic Data

> This is planning guidance, **not legal advice**. Confirm each step with the faculty advisor, the school's IRB, and the clinic's privacy officer. Contacts were checked against public websites and directories in November 2026. Confirm them before use.

1. What "real data" means for this project

The screening and referral logs this project measures **do not exist yet**, so no one can send them. They are created when a clinic uses the protocol. Public data can describe the community, but it cannot measure this clinic's screening or referral rates (P1–P4 in 07).

Data	Exists today?	How to get it	Replaces synthetic data?
Screening and referral logs	No. Created by running the protocol	Clinic partner + approvals (§2)	**Yes**
Baseline chart audit	Yes, in the clinic's records	Clinic partner + approvals (§2)	**Yes** (baseline weeks)
Public aggregate data (homeless count, health centers, food insecurity)	Yes	Download (§4)	No. Background, comparison, and simulation inputs only
Resource directory checks	Can be collected now	Phone calls (§7)	No. Supports 03 and 04

2. Path to real clinic data

Do not open patient records until steps 1–5 are done.

#	Step	Who	Result	See
1	Find a clinic partner and sponsor	Student	Letter or email of support	§5, Template A
2	Advisor approves the plan	Faculty advisor	Approval	00, 06
3	QI vs. research determination	School IRB	Determination letter or IRB approval	08 §1
4	Clinic onboarding	Clinic HR or compliance	Student or volunteer agreement, HIPAA training, confidentiality agreement	08 §2
5	Choose how data leaves the clinic	Clinic privacy officer	Written choice of route (§3)	08 §2
6	Adapt and verify materials	Student and sponsor	Resource directory verified; safety pathway signed off	03, 04
7	Baseline chart audit (weeks 1–4)	Student, on clinic systems	Baseline rows in `screening_log.csv`	07

#	Step	Who	Result	See
8	Go live with PDSA cycles	Clinic staff	Screening and referral logs, collected as visits happen	06
9	Analyze and report	Student or clinic staff	Aggregate results only	§3

Steps 1–5 can take weeks to months. Start partner calls and the IRB question early.

3. Handling the real data

Coded logs are not de-identified. They contain dates of service, so they are PHI under HIPAA even without names. Pick one route with the privacy officer (08 §2):

Route	Who handles row-level data	Paperwork	What leaves the clinic
1. Analyze on site	Student, as a clinic workforce member, on clinic-controlled storage	Student or volunteer agreement; HIPAA training	Aggregate results
2. Clinic runs the analysis	Clinic staff only	Privacy officer reviews outputs	Aggregate results
3. Limited Data Set	Student, off site	Data Use Agreement	Coded logs with dates

Run the analysis on the approved machine:

```
mkdir -p data/real                    # git-ignored
cp data/templates/*.csv data/real/    # blank logs with the right headers (or save clinic exports here)
# enter or export the data, then:
python analysis/analyze.py --data-dir data/real --out-dir data/real/output --strict
git status --short                    # data/real must not appear
```

- `--strict` stops on values that don't match the data dictionary. Fix the log and rerun.
- Keep `data/synthetic/` and `output/` as the demo. Never overwrite them with real data.
- Never commit anything from `data/real/`. Don't move it through cloud sync, email, or personal devices unless the clinic approves.
- `weekly_measures.csv` and the run charts contain unsuppressed weekly counts. The privacy officer reviews them before they leave the clinic.

4. Data that is already available

These sources are free and public, and need no approval. Use them for the background section, for comparisons, and to ground the simulation's assumptions. Do **not** present them as this project's results.

Source	What it gives	Where
Fairfax County Point-in-Time Count 2026 (OPEH)	1,365 people experiencing homelessness on Feb 4, 2026 (up 3% from 1,322 in 2025); demographics; shelter and housing inventory	fairfaxcounty.gov/homeless/point-time-count-2026

Source	What it gives	Where
HRSA Uniform Data System (UDS)	National, state, and health-center data, including patients experiencing homelessness. Starting with 2026 data, Table 6A adds counts of patients screened for upstream drivers of health and patients with food insecurity, housing instability, or financial insecurity. The first public release is likely in mid-2027	data.hrsa.gov
CDC PLACES (December 2025 release)	County and census-tract model estimates of 7 health-related social needs, including food insecurity, housing insecurity, and transportation barriers. Virginia is covered (it used the BRFSS social needs module in 2023). CDC cautions against using PLACES to evaluate programs	cdc.gov/places
Feeding America, Map the Meal Gap	County food insecurity estimates (data through 2024, released July 2026); dataset request form	map.feedingamerica.org
Capital Area Food Bank, Hunger Report 2026	NORC survey of 3,804 adults (May 2026): 27% of Fairfax County households food insecure; 38% across the region	capitalareafoodbank.org

General-population rates likely understate needs among people experiencing homelessness. If you use them in the simulation, say so and cite the source.

5. Who to contact

Possible clinic partners

Organization	Why	Phone	Email
New Hope Housing, Mobile Medical Outreach	Mobile care for people experiencing homelessness; closest match	703-799-2293	admin@newhopehousing.org
Fairfax County Health Department, Homeless Healthcare Program	Nurse practitioners in shelters, drop-in centers, and street outreach	703-246-2411 (main line; ask for the program)	
GMU MAP Clinics	Academic clinics, including shelter-based sites	703-993-5880	mapclinic@gmu.edu
Neighborhood Health (FQHC)	Health center; may share aggregate screening counts	703-535-5568	partners@nhnva.org
HealthWorks for Northern Virginia (FQHC)	Health center with a Sully site (13800 Wall Rd, Herndon)	703-443-2000	

Data holders

Organization	Ask for	Phone	Email / mail
Fairfax County Office to Prevent and End Homelessness (OPEH)	Aggregate Point-in-Time or HMIS figures beyond the dashboard (Template B)	703-246-5000	OPEHGeneralMail@fairfaxcounty.gov · 3700 Pender Drive, Fairfax, VA 22030
Fairfax County FOIA Office	Existing aggregate records only (Template C)	703-324-7052	FOIA@fairfaxcounty.gov · 12000 Government Center Pkwy, Suite 551, Fairfax, VA 22035
HRSA UDS Support Center	Help reading UDS data	866-UDS-HELP (866-837-4357)	udshelp330@bphcdata.net
Feeding America	Map the Meal Gap dataset		Online request form (feedingamerica.org, Map the Meal Gap)
Your school's IRB or QI office	QI determination (step 3)		

Questions to ask a clinic

1. Do you document on paper or in an EHR? Which one, and does it work offline?
2. Do you screen for social needs now? Which tool, who asks, and how often?
3. How many unique adult patients do you see per week, and how many visits?
4. Are you an FQHC or Health Care for the Homeless grantee that reports UDS?
5. How do you track referrals and follow-up?
6. Would you sponsor a student QI project? What onboarding is required?

6. Outreach templates

Replace everything in [brackets].

Template A — Clinic partnership (email or letter)

> **Subject:** Student quality improvement project on social needs screening
>
> Dear [Name],
>
> I am a [program] student at [school], supervised by [advisor name, title]. For my quality improvement project, I have prepared a ready-to-use package for standard social needs screening at a mobile clinic serving people experiencing homelessness. It includes a short questionnaire based on the CMS Accountable Health Communities tool, a screening protocol, staff training, a Fairfax County resource directory, and a measurement plan.
>
> I am looking for a clinic partner to pilot it for about [16] weeks. The project would:
>
> - Screen adult patients for housing, food, transportation, utilities, safety, health literacy, and ID needs, and offer referrals.
> - Measure the share of eligible visits screened and referred, using coded logs that stay under the clinic's control.

> • Start only after my school's QI determination and your clinic's approval.
>
> Could we set up a 15-minute call? I would also like to learn how your clinic documents social needs today.
>
> Thank you, [Name] · [phone] · [email]

Template B — Aggregate data request

> **Subject:** Request for aggregate data for a student QI project
>
> Dear [Name],
>
> I am a [program] student at [school] working on a quality improvement project about social needs screening for adults experiencing homelessness in Fairfax County. I am asking for **aggregate figures only**, with no patient-level data, for [time period], if they already exist:
>
> 1. [Number of adult patients or visits]
> 2. [Number screened for social needs, and the tool used]
> 3. [Number who screened positive for food, housing, or transportation needs]
> 4. [Number referred]
>
> Counts under [5] may be suppressed. I will use the figures only as background for my [course project/ thesis] and will cite them as you direct.
>
> Thank you, [Name] · [phone] · [email] · Advisor: [name, email]

Template C — Virginia FOIA request (existing records)

> **To:** Fairfax County FOIA Office, FOIA@fairfaxcounty.gov
>
> Under the Virginia Freedom of Information Act (Va. Code § 2.2-3700 et seq.), I request copies of the following existing public records: [for example, the most recent annual report or performance summary for the Health Department's Homeless Healthcare Program showing aggregate numbers of people served]. I am a Virginia resident. I prefer electronic copies. Please tell me in advance if costs will exceed $[amount].
>
> [Name] · [mailing address] · [email]

Virginia FOIA covers existing records only. Agencies do not have to create new reports, and patient health records are exempt. Requesters must be Virginia citizens or qualifying media.

7. If no partner is available in time

- Keep the synthetic data and label it **simulated** everywhere.
- Replace the example probabilities in `analysis/generate_synthetic_data.py` with sourced estimates, and cite each one. Prefer published studies of patients experiencing homelessness, and use the §4 data for local context.

- Ask the advisor whether an implementation-ready package with simulated results meets the course requirements.
- Collect real data that involves no patients now. Call every resource in 04, confirm hours, eligibility, and intake steps, and record the date in the “Verified” column.

8. Checklist

- [ ] Clinic partner and sponsor confirmed
- [ ] Advisor approval
- [ ] QI/IRB determination letter on file
- [ ] Clinic onboarding complete
- [ ] Route for getting data out of the clinic chosen in writing
- [ ] Resource directory verified by phone
- [ ] Baseline audit complete (weeks 1–4)
- [ ] Analysis run in `data/real/` on approved storage; outputs reviewed by the privacy officer

Data Dictionary

Two CSV logs with one shared schema. They are used for synthetic data (`data/synthetic/`), blank templates (`data/templates/`), and real coded data (`data/real/`, **git-ignored, never commit**).

No direct identifiers in these files: no names, DOB, addresses, phone numbers, or free-text clinical detail. `patient_id` is a study code (e.g., `P001`). The linking key stays at the clinic. Real logs are still **PHI** because they contain dates of service. See 08 §2 before any data leaves the clinic.

`screening_log.csv` — one row per adult visit

Field	Type	Allowed values	Notes
`visit_id`	text	unique, e.g. `V0001`	
`patient_id`	text	e.g. `P001`	Study code
`visit_date`	date	`YYYY-MM-DD`	HIPAA date element; see 08 §2 before sharing
`week`	int	1..N	Project week (week 1 = first baseline week); derived from dates, so still a date element
`phase`	text	`baseline`, `pdsa1`, `pdsa2`, `pdsa3`, `sustain`	
`excluded`	Y/N	`Y`, `N`	Y = medical/unable; removed from denominators
`exclusion_reason`	text	blank, `medical_unstable`, `unable_to_participate`	Required if `excluded = Y`
`screen_status`	text	`complete`, `partial`, `declined`, `not_screened`, `not_due`	`not_due` = complete screen within prior 90 days
`screener_role`	text	`MA`, `RN`, `CHW`, `clinician`, `volunteer`, blank	Blank if not screened
`language`	text	`en`, `es`, `other`	
`minutes`	int	blank or ≥ 0	Time to administer (balancing B2)
`housing_status`	text	`steady`, `at_risk`, `unstable`, blank	Q1
`sleep_location`	text	`unsheltered`, `vehicle`, `shelter`, `doubled_up`, `motel`, `own_place`, `other`, blank	Q1a
`housing_quality_pos`	text	`Y`, `N`, `NA`, blank	Q2; `NA` if no steady place
`food_pos`	text	`Y`, `N`, blank	Q3–4
`transport_pos`	text	`Y`, `N`, blank	Q5
`utilities_pos`	text	`Y`, `N`, `NA`, blank	Q6; `NA` if no steady place
`safety_pos`	text	`Y`, `N`, `deferred`, blank	Q7–10 sum ≥ 11

Field	Type	Allowed values	Notes
`literacy_pos`	text	`Y`, `N`, blank	Q11 (clinical response, not a referral)
`id_docs_pos`	text	`Y`, `N`, blank	Q12
`notes`	text	short, **no identifiers or clinical detail**	e.g. `baseline: food asked only`

Domain fields are filled when `screen_status` is `complete` or `partial`. During baseline, `partial` records which domains happened to be asked.

`referral_log.csv` — one row per domain referral offered

Field	Type	Allowed values	Notes
`referral_id`	text	unique, e.g. `R0001`	
`visit_id`	text	must exist in screening log	
`patient_id`	text		
`referral_date`	date	`YYYY-MM-DD`	
`domain`	text	`housing`, `food`, `transportation`, `utilities`, `safety`, `id_docs`	Housing quality → `housing`
`accepted`	Y/N	`Y`, `N`	N = offered, patient declined
`referral_type`	text	`warm_handoff`, `direct_referral`, `information_only`	
`resource`	text	short name from the resource directory	e.g. `WFCM`, `CSP`
`followup_date`	date	blank or `YYYY-MM-DD`	
`followup_status`	text	`pending`, `connected`, `not_connected`, `unable_to_reach`, `declined_followup`	

Every row is an *offered* referral. “Offered for every actionable positive domain” (P2) is checked by matching domains to the visit’s positive screen fields.

Derived variables (computed in `analysis/analyze.py`)

- `housing_pos` = `housing_status` ∈ {`at_risk`, `unstable`}
- `due` = no `complete` screen for that patient in the 90 days before `visit_date`
- `up_to_date` = `complete` at this visit, or a `complete` screen within the prior 90 days
- `actionable_positive_domains` = positive among housing (status or quality), food, transportation, utilities, safety, id_docs

Social Needs Screening Form — v0.1

Study ID: _____ **Date:** ____/____/_____ **Screener role:** ☐ MA ☐ RN ☐ CHW ☐ Clinician ☐ Volunteer **Language:** ☐ English ☐ Spanish ☐ Other: _____ (interpreter used? ☐ Y ☐ N) **Start time:** ____ **End time:** ____

☐ Patient declined screening (offer resource card) ☐ Not due (complete screen within 90 days)

> *Read aloud:* "We ask everyone we see the same few questions about things like food, transportation, and safety, because these affect health. Your answers are private and won't affect your care today. You can skip any question. If something comes up, we can connect you with help, only if you want it."

---

1. What is your living situation today?

- ☐ I have a steady place to live
- ☐ I have a place to live today, but I am worried about losing it in the future **[+]**
- ☐ I do not have a steady place to live (temporarily staying with others, in a hotel, in a shelter, living outside on the street, on a beach, in a car, abandoned building, bus or train station, or in a park) **[+]**

1a. Where did you sleep last night? ☐ Outside ☐ Vehicle ☐ Shelter/hypothermia site ☐ With friends/family ☐ Motel ☐ Own place ☐ Other: ____

2. *(Skip if no steady place.)* **Think about the place you live. Do you have problems with any of the following?** ☐ Pests ☐ Mold ☐ Lead paint or pipes ☐ Lack of heat ☐ Oven/stove not working ☐ Smoke detectors missing/not working ☐ Water leaks ☐ None **[+ if any]**

3. Within the past 12 months, you worried that your food would run out before you got money to buy more. ☐ Often true **[+]** ☐ Sometimes true **[+]** ☐ Never true

4. Within the past 12 months, the food you bought just didn't last and you didn't have money to get more. ☐ Often true **[+]** ☐ Sometimes true **[+]** ☐ Never true

5. In the past 12 months, has lack of reliable transportation kept you from medical appointments, meetings, work, or from getting things needed for daily living? ☐ Yes **[+]** ☐ No

6. *(Skip if no steady place.)* **In the past 12 months has the electric, gas, oil, or water company threatened to shut off services in your home?** ☐ Yes **[+]** ☐ No ☐ Already shut off **[+]**

Safety: ask ONLY if private. ☐ Deferred: no privacy

How often does anyone, including family and friends:	**Never 1**	**Rarely 2**	**Sometimes 3**	**Fairly often 4**	**Frequently 5**
7. physically hurt you?	☐	☐	☐	☐	☐

How often does anyone, including family and friends:	Never 1	Rarely 2	Sometimes 3	Fairly often 4	Frequently 5
8. insult or talk down to you?	☐	☐	☐	☐	☐
9. threaten you with harm?	☐	☐	☐	☐	☐
10. scream or curse at you?	☐	☐	☐	☐	☐

Total (7–10): ____ **[+ if ≥ 11 → safety pathway]**

11. How often do you need to have someone help you when you read instructions, pamphlets, or other written material from your doctor or pharmacy? ☐ Never ☐ Rarely ☐ Sometimes **[+]** ☐ Often **[+]** ☐ Always **[+]** → *use teach-back*

12. Do you have these documents (with you or somewhere safe you can get to)?

	Have	Don't have [+]	Want help?
Photo ID	☐	☐	☐ Y ☐ N
Social Security card	☐	☐	☐ Y ☐ N
Birth certificate	☐	☐	☐ Y ☐ N
Insurance/Medicaid card	☐	☐	☐ Y ☐ N

13. Would you like help today with: ☐ Housing ☐ Food ☐ Transportation ☐ Utilities ☐ Safety ☐ ID/documents ☐ None right now

14. Best way to reach you in the next few weeks: ☐ Phone/text: _____ ☐ Shelter/case manager: _____ ☐ Next clinic stop ☐ No follow-up

---

Staff scoring box

Domain	+	−	N/A / deferred	Referral offered?	Accepted?	Type (WH/DR/INFO)	Resource
Housing	☐	☐		☐	☐		
Food	☐	☐		☐	☐		
Transportation	☐	☐		☐	☐		
Utilities	☐	☐	☐	☐	☐		
Safety	☐	☐	☐	☐	☐		
Health literacy	☐	☐		*clinical: teach-back*			
ID/documents	☐	☐		☐	☐		

Logged to screening_log ☐ referral_log ☐ Initials: ____

Items 1–10: CMS Accountable Health Communities HRSN Screening Tool. Item 11: Single Item Literacy Screener (Morris et al., 2006). Items 1a, 12–14: locally developed.

PDSA Worksheet

Cycle #: ___ **Change being tested:** _____ **Dates:** _____ to _____
Owner: _____ **Linked driver** (06): _____

PLAN

- **Objective of this cycle:**
- **Questions:**
- **Predictions** (for each question, include a number):
- **Plan:** who / what / where / when
- **Data to collect** (measure IDs from 07):

DO

- What happened? Was the plan carried out as intended?
- Problems or unexpected observations:
- Data collected:

STUDY

- Compare the results with the predictions (attach the run chart):
- What did we learn?

ACT

☐ **Adopt** ☐ **Adapt** ☐ **Abandon** - Changes for the next cycle: - Next cycle starts: _____

QI Measure Summary

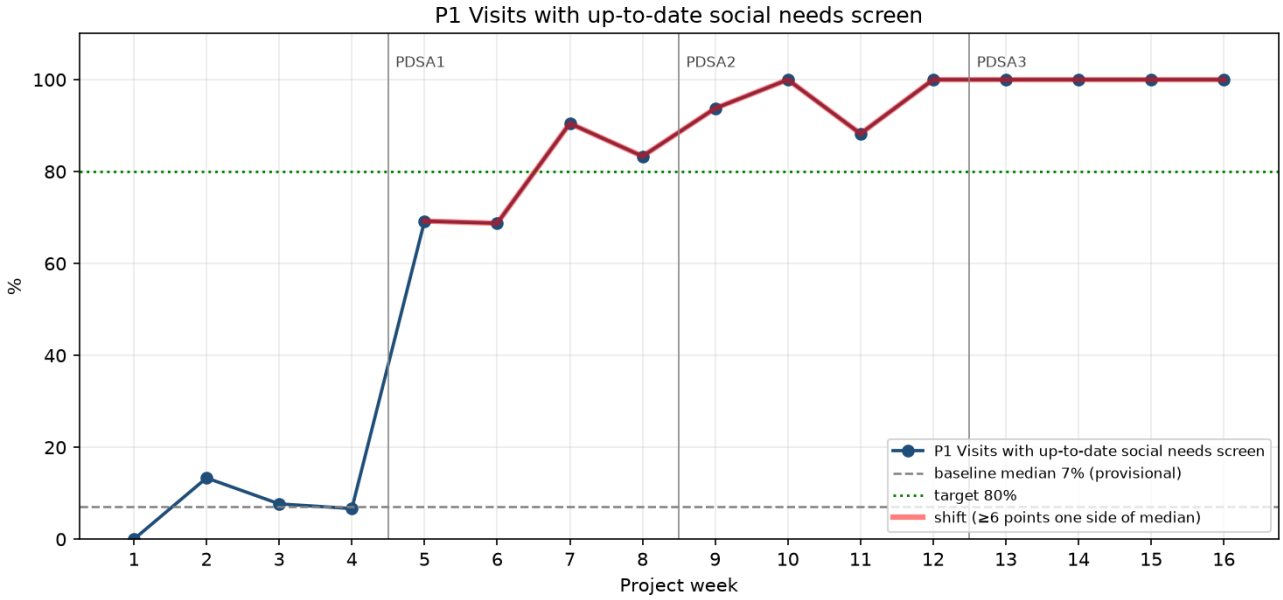
> ⚠️ **SYNTHETIC DATA — for demonstration only. Not real patients; do not cite.**

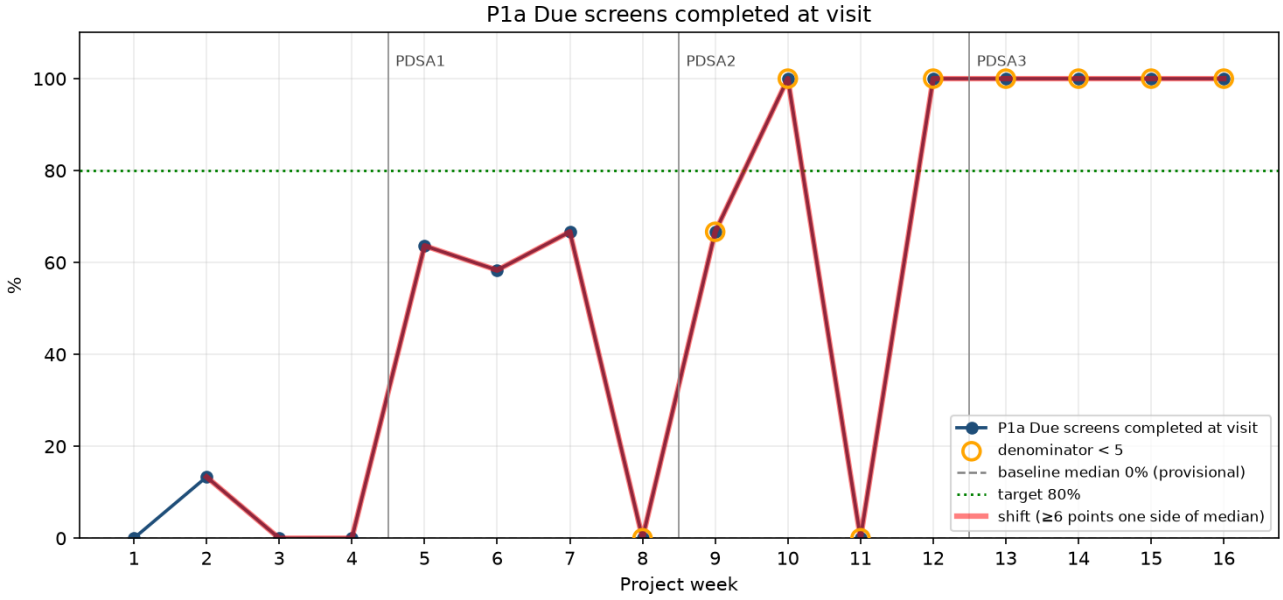
Data: `data/synthetic` · visits: 245 (4 excluded) · patients: 30 · weeks: 1–16 · referrals: 65

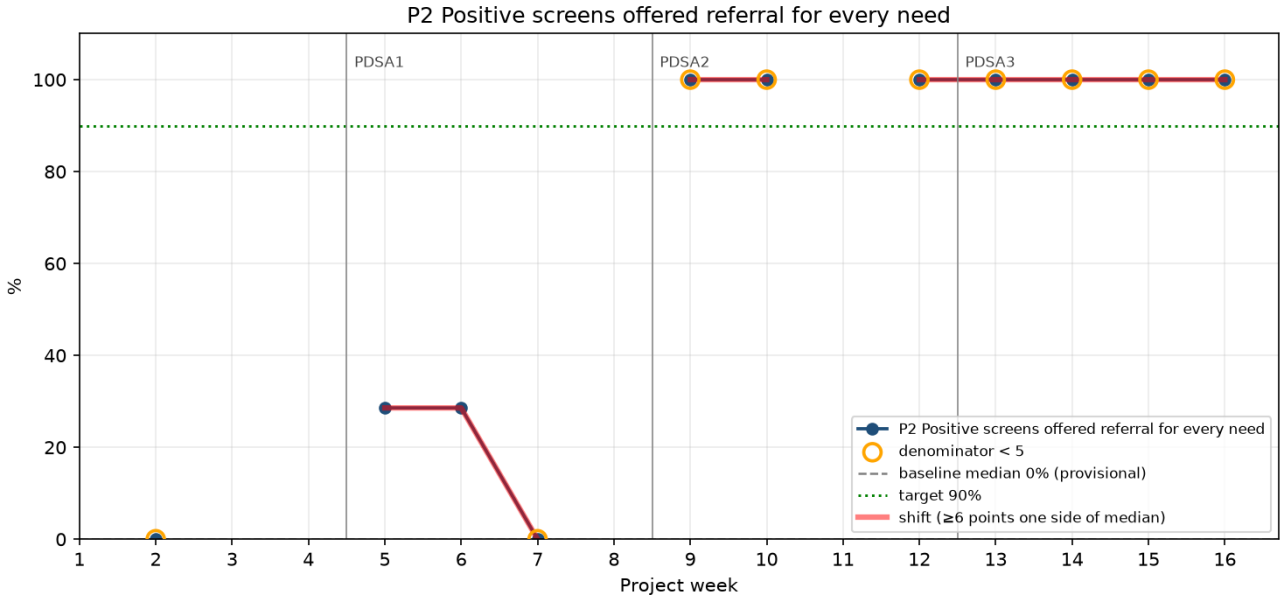
Validation issues: 0

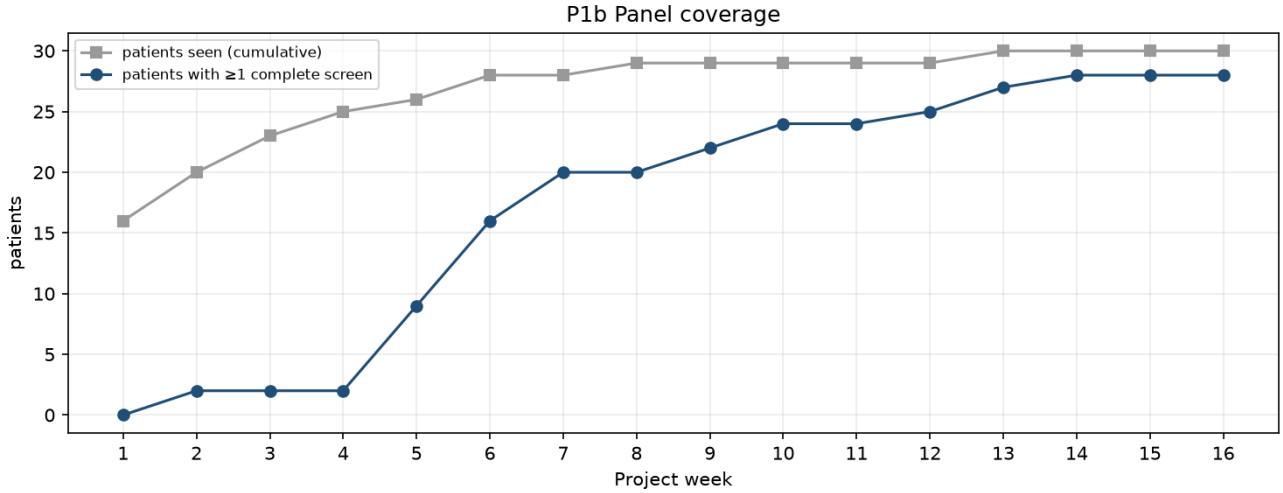
Run chart signals (Perla et al. 2011 rules)

- **P1** (baseline median 7%): shift above median, weeks 5–16
- **P1a** (baseline median 0%): shift above median, weeks 2–16
- **P2** (baseline median 0%): shift above median, weeks 5–16


P1 Visits with up-to-date social needs screen
PDSA1
PDSA2
PDSA3
%
Project week
P1 Visits with up-to-date social needs screen
baseline median 7% (provisional)
target 80%
shift (≥6 points one side of median)


P1a Due screens completed at visit
PDSA1
PDSA2
PDSA3
%
Project week
P1a Due screens completed at visit
denominator < 5
baseline median 0% (provisional)
target 80%
shift (≥6 points one side of median)





Baseline vs. intervention (descriptive, Wilson 95% CI)

measure	period	num	den	pct	95% CI
P1 up-to-date screen	baseline	4	63	6%	2%–15%
P1a due-screen completed	baseline	2	61	3%	1%–11%
P2 referral offered (all positive domains)	baseline	0	2	0%	0%–66%
B1 declined (due visits)	baseline	1	61	2%	0%–9%
P1 up-to-date screen	intervention	162	178	91%	86%–94%

measure	period	num	den	pct	95% CI
P1a due-screen completed	intervention	28	44	64%	49%–76%
P2 referral offered (all positive domains)	intervention	14	28	50%	33%–67%
B1 declined (due visits)	intervention	4	44	9%	4%–21%

B2 Screening time: median 7 min (IQR 6–8), n=30

Follow-up (accepted referrals ≥ 30 days old, grouped by phase of referral)

phase	accepted	documented	connected	O1	O2
baseline	8	1	1	12%	12%
pdsa1	23	6	5	26%	22%
pdsa2	11	4	2	36%	18%
all	42	11	8	26%	19%

Referral funnel

step	count
Positive actionable domains (complete screens)	75
Referrals offered	57
Referrals accepted	45
Follow-up documented (any time)	12
Connected	8

Need prevalence (latest complete screen per patient)

domain	patients	of	pct
Housing unstable/at risk	25	28	—
Housing quality problem (of those with housing)	<5	4	—
Food	22	28	79%
Transportation	14	28	50%
Utilities (of those with housing)	<5	4	—
Interpersonal safety (of those asked privately)	<5	22	—
Safety deferred (no privacy)	6	28	21%
Health literacy	10	28	36%
ID/documents	5	28	18%

P1b panel coverage at final week: 28/30 patients.

Weekly data

week	phase	visits	P1	due	P1a	p2_den	P2	B1
1	baseline	20	0%	20	0%	0	—	0%
2	baseline	15	13%	15	13%	2	0%	0%
3	baseline	13	8%	12	0%	0	—	0%
4	baseline	15	7%	14	0%	0	—	7%
5	pdsa1	13	69%	11	64%	7	29%	9%
6	pdsa1	16	69%	12	58%	7	29%	17%
7	pdsa1	21	90%	6	67%	4	0%	17%
8	pdsa1	12	83%	2	0%	0	—	0%
9	pdsa2	16	94%	3	67%	2	100%	0%
10	pdsa2	9	100%	2	100%	2	100%	0%
11	pdsa2	17	88%	2	0%	0	—	0%
12	pdsa2	16	100%	1	100%	1	100%	0%
13	pdsa3	14	100%	2	100%	2	100%	0%
14	pdsa3	17	100%	1	100%	1	100%	0%
15	pdsa3	13	100%	1	100%	1	100%	0%
16	pdsa3	14	100%	1	100%	1	100%	0%